

11	C	New GPE = $1.00 + mgh = 1.00 + (0.200)(0.3) = 1.06 \text{ J}$
12	D	From $pV = \frac{1}{3} Nm \langle c^2 \rangle$, $p = \frac{1}{3} \rho \langle c^2 \rangle$, $(1.00 \times 10^5 \text{ Pa}) = \frac{1}{3} (1.60 \text{ kg m}^{-3}) \langle c^2 \rangle$, $\sqrt{\langle c^2 \rangle} = 433 \text{ m s}^{-1}$
13	B	Arrow from B to A is longer than from A to B. This means that more thermal energy is transformed from B to A, i.e. temperature of B is higher than A.